



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru

BTech VI Semester

End Semester Examination - April / May -2025

BUSINESS INTELLIGENCE AND ANALYTICS [CSE-4018]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Compare different types of Business Analytics with suitable real-world examples, and analyze how each type can support strategic decision-making in such an organization. (5)
 - A)
 - B) Given a retail business experiencing challenges in understanding customer behavior and managing large volumes of transaction data, Illustrate how can Big Data Analytics be applied to improve decision-making and operational efficiency. (5)
- 2) Demonstrate different phases of Information Hierarchy graph in Business Intelligence using a neat diagram. (5)
 - A)
 - B) A multinational e-commerce company wants to enhance its decision-making process using Business Intelligence (BI) dimensions. The company deals with a large volume of data related to sales, customer behaviour, and inventory management across multiple regions. (5)

Examine two important BI dimensions that the company should consider for analysing its business performance and its contributions to strategic decision making.
- 3) With a neat diagram explain how does a value-driven approach influence BI implementation and how organizations derive value from BI by effective information usage. (5)
 - A)
 - B) A retail chain requires a real-time dashboard for monitoring in-store purchases and online orders to adjust inventory dynamically. Illustrate with a neat diagram the high architectural components of the BI system would need to be updated to support real-time data processing and visualization. (5)
- 4) Assume a fast-growing retail company wants to implement RFID. Discuss with the neat diagram the working process of RFID tags in the retail company with its advantages. (5)
 - A)
 - B) A company is using customer data to improve its marketing strategy through Business Intelligence tools. Apply the concepts of privacy and ethics to this scenario underling the ethical considerations to be taken into account, ensuring responsible data usage. (5)
- 5) A large e-commerce company wants to analyse customer behaviour using Big Data analytics. They need to discover patterns in purchasing trends, identify high-value customers, and detect fraud transactions. Since they handle large volumes of data, they (5)

- A) prefer to use open-source technologies for cost-effectiveness and scalability.
Compare any two open-source Big Data technologies with its role in gaining insights into customer behaviour and detect fraud transactions.
- B) A retail company collects massive amounts of customer transaction data. They want to use Big Data analytics to understand customer buying behaviour and improve sales.
List any five predictive analytics methods/approaches used to recommend products with (5) its role in Big Data framework to process large scale data.

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